Proposed Solution

Date	17 September 2026
Team ID	SWTID-2026-2630
Project Name	AI Fit Track API
Maximum Marks	2 Marks

Proposed Solution

Element	Description
Project Title	AI Fit Track – AI-Powered Workout Tracking & Fitness Coaching Backend
Objective	Build a secure, RESTful Node.js/Express backend that lets an authenticated user manage their workout history and receive AI-generated (Google Gemini) workout recommendations and fitness insights derived from that history.
Scope	User registration/login (JWT + bcrypt), full CRUD and search on Workout records, and two Gemini-backed AI endpoints (recommendation and insights). A `client/` folder is scaffolded for a future UI but a dedicated frontend is out of scope for this cycle – the API is validated through Thunder Client / Postman.
Problem it Solves	Removes fragmented manual logging (notebooks/spreadsheets), removes manual-entry data errors via schema validation, and adds a personalized AI guidance layer that no manual method can offer.
Proposed Solution / Approach	A modular MVC-style Express application (`config / controllers / middleware / models / routes / services`) backed by MongoDB via Mongoose, JWT-protected routes, and a dedicated `geminiService.js` that isolates all Gemini AI calls from the rest of the business logic.
Resource Requirements	Node.js v16+, Express.js, MongoDB (local instance, `mongodb://localhost:27017/aifittrack`), Mongoose, `jsonwebtoken` , `bcryptjs` , Google Gemini API key, VS Code + Thunder Client/Postman, Git & GitHub, 5-member team over the 15–29 September 2026 cycle.